

## 2.05 Statistical Hypothesis Testing

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| 2.05a | The language of hypothesis testing | <p>a) Understand and be able to use the language of statistical hypothesis testing, developed through a binomial model: null hypothesis, alternative hypothesis, significance level, test statistic, 1-tail test, 2-tail test, critical value, critical region, acceptance region, <math>p</math>-value.</p> <p><i>Hypotheses should be stated in terms of parameter values (where relevant) and the meanings of symbols should be stated. For example,</i><br/>“<math>H_0: p = 0.7</math>, <math>H_1: p \neq 0.7</math>, where <math>p</math> is the population proportion in favour of the resolution”.</p> <p><i>Conclusions should be stated in such a way as to reflect the fact that they are not certain. For example,</i><br/>“There is evidence at the 5% level to reject <math>H_0</math>. It is likely that the mean mass is less than 500 g.”<br/>“There is no evidence at the 2% level to reject <math>H_0</math>. There is no reason to suppose that the mean journey time has changed.”</p> <p><i>Some examples of incorrect conclusion are as follows:</i><br/>“<math>H_0</math> is rejected. Waiting times have increased.”<br/>“Accept <math>H_0</math>. Plants in this area have the same height as plants in other areas.”</p> | MO1 |
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| OCR Ref.           | Subject Content                                               | Stage 1 learners should ...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Stage 2 learners additionally should ... | DfE Ref. |
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| 2.05b<br><br>2.05c | Hypothesis test for the proportion in a binomial distribution | <p>b) Be able to conduct a statistical hypothesis test for the proportion in the binomial distribution and interpret the results in context.</p> <p>c) Understand that a sample is being used to make an inference about the population and appreciate that the significance level is the probability of incorrectly rejecting the null hypothesis.</p> <p><i>Learners should be able to use a calculator to find critical values.</i></p> <p><i>Includes understanding that, where the significance level of a test is specified, the probability of the test statistic being in the rejection region will always be less than or equal to this level.</i></p> <p><i>[The use of normal approximation is excluded.]</i></p> |                                          | MO2      |



| OCR Ref.           | Subject Content                                         | Stage 1 learners should ... | Stage 2 learners additionally should ...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | DfE Ref. |
|--------------------|---------------------------------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 2.05f<br><br>2.05g | Hypothesis test using Pearson's correlation coefficient |                             | <p>f) Understand Pearson's product-moment correlation coefficient as a measure of how close data points lie to a straight line.</p> <p>g) Use and be able to interpret Pearson's product-moment correlation coefficient in hypothesis tests, using either a given critical value or a <math>p</math>-value and a table of critical values.</p> <p><i>When using Pearson's coefficient in an hypothesis test, the data may be assumed to come from a bivariate normal distribution.</i></p> <p><i>A table of critical values of Pearson's coefficient will be provided.</i></p> <p><i>[Calculation of correlation coefficients is excluded.]</i></p> | MO1      |

